

ou 2m

Chapter 2 and 1 Test T/I (2023V1)

Name: [REDACTED]

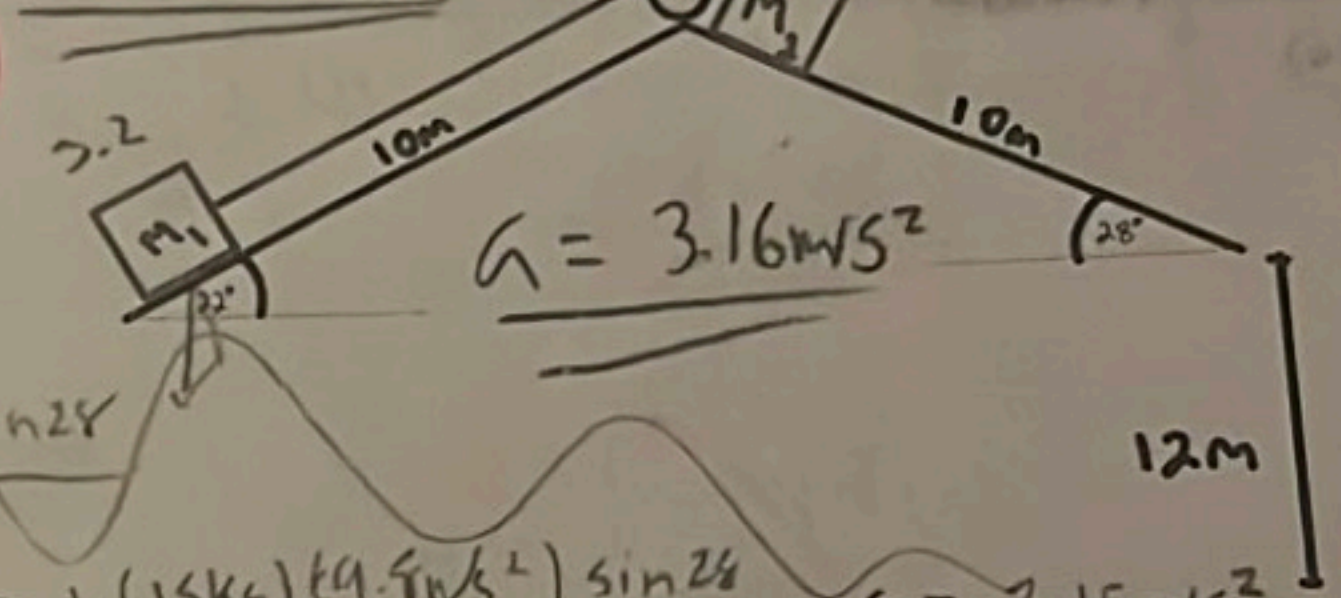
16.5 / 20

For full marks, Clearly show ALL of your work for every question. Include FBD's, directions where appropriate. Round your final answer to 3 sig digs.

1. Two blocks are connected to each other through a pulley as shown in the diagram below. The length of each ramp is 12.0 m. Block 1 has a mass of 3.2 kg and block 2 has a mass of 15 kg. Assume no friction.

- a. What is the acceleration of the system and the tension in the rope? (3 marks)

$$F_T = 21.9 \text{ N}$$



$$a = 3.16 \text{ m/s}^2$$

$$a = \frac{-(3.2 \text{ kg})(9.8 \text{ m/s}^2) \sin 22^\circ + (15 \text{ kg})(9.8 \text{ m/s}^2) \sin 28^\circ}{18.2 \text{ kg}} = 3.15 \text{ m/s}^2$$

- b. Assuming the rope breaks immediately when block 2 reaches the end of the ramp, what is the range of block 2 (from the end of the ramp) assuming it becomes a projectile with no air resistance? (5 marks)

Done on sep sheet of paper.

$$Ad = 12.0 \text{ m}$$



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3a) $m_1 a + m_2 a = m_1 g - m_2 g \sin 50 - m_1 g \cos 50 \mu_k$
 $a (m_1 + m_2) = (6.5 \text{ kg}) - (4.5 \text{ kg})(9.8 \text{ m/s}^2) \sin 50 - (4.5 \text{ kg})(9.8 \text{ m/s}^2) \cos 50 (0.14)$
 $a = \frac{63.7 - 33.8}{11 \text{ kg}} = 3.97$
 $a = 2.36 \text{ m/s}^2$

$$m_1 a = m_1 g - F_T$$

$$F_T = m_1 g - m_1 a$$

$$= (6.5 \text{ kg})(9.8 \text{ m/s}^2) - (6.5 \text{ kg})(2.36 \text{ m/s}^2)$$

$$F_T = 63 - 15.34$$

$$F_T = 48.0 \text{ N}$$

b) $F_s = F_N \mu_k$

$$\frac{\sin 50 (m_1 g)}{\cos 50 (m_1 g)} = \frac{\cos 50 m_1 g \mu_k}{\cos 50 m_1 g}$$

$$1.19 = \mu_k$$

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1a

$$F = m_1 a$$

$$F = F_T - F_g$$

$$= F_T - m_1 g \sin 22$$

$$F = m_2 L$$

$$F = F_g - F_T$$

$$= m_2 g \sin 28 - F_T$$

$$m_1 a + m_2 a = -m_1 g \sin 22 + m_2 g \sin 28$$

$$a = \frac{-m_1 g \sin 22 + m_2 g \sin 28}{18.2 \text{ kg}}$$

$$a = \frac{-(3.2)(9.8) \sin 22 + (15)(9.8) \sin 28}{18.2 \text{ kg}}$$

$$a = \frac{-11.75 + 64}{18.2 \text{ kg}}$$

$$14.6 \text{ kg}$$

$$a = 3.16 \text{ m/s}^2$$

$$m_1 a = F_T - m_1 g \sin 22$$

$$m_1 a + m_1 g \sin 22 = F_T$$

$$21.9 \text{ N} = F_T$$

1b

$$d = 12 \text{ m}$$

$$a = 3.16$$

$$t =$$

$$d = ? \quad t = 2.76 \text{ s}$$

$$\Delta d = v_i \Delta t + \frac{1}{2} a \Delta t^2$$

$$12 \text{ m} = \frac{1}{2} (3.16 \text{ m/s}^2) t^2$$

$$\sqrt{\frac{12 \text{ m}}{1.58}} = \sqrt{t^2}$$

$$t = 2.76 \text{ s}$$

$$\Delta d = \frac{1}{2} a \Delta t^2$$

$$= \frac{1}{2} (3.16 \text{ m/s}^2) (2.76)^2$$

$$\Delta d = 12.0 \text{ m}$$

Julia Silvira Delacruz

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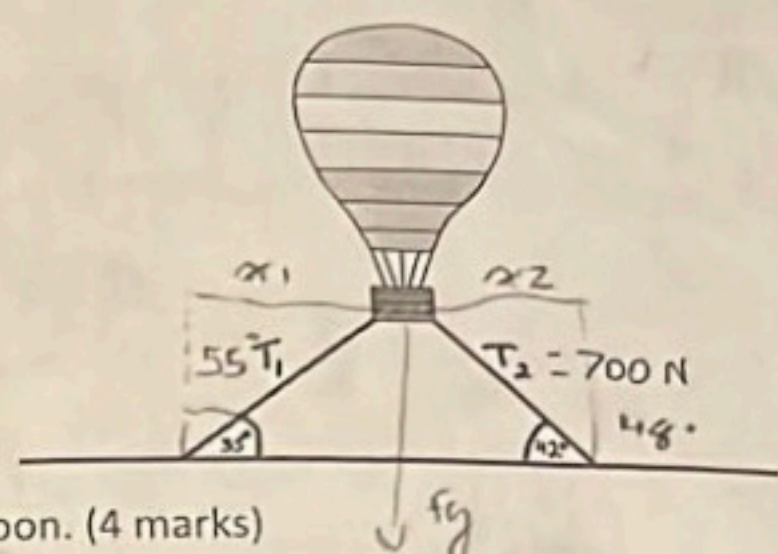
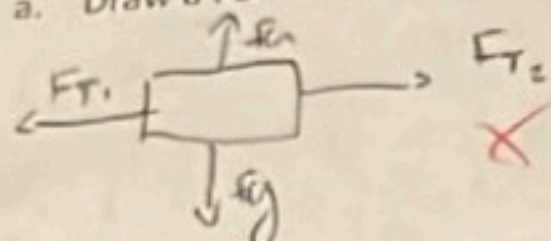


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2. Consider the diagram below at static equilibrium. (6 marks)

a. Draw a FBD for the forces acting on the balloon. (2 marks)



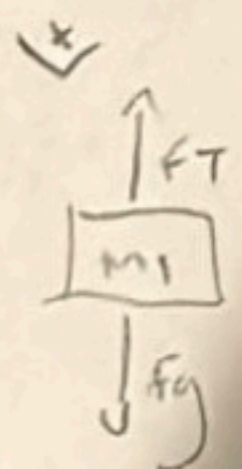
b. Calculate the tension in each rope and the applied force on the balloon. (4 marks)

$$\begin{aligned} \textcircled{1} \quad T_1 \sin 55^\circ &= T_2 \sin 48^\circ \\ \sin 55^\circ T_1 &= \frac{\sin 48^\circ T_2}{\sin 55^\circ} \\ T_1 &= \frac{520 \text{ N}}{\sin 55^\circ} \\ T_1 &= 635 \text{ N} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad F_{y1} + F_{y2} + F_g &= 0 \\ F_{y1} + F_{y2} &= -F_g \\ \cos 55^\circ (635 \text{ N}) + \cos 48^\circ (700 \text{ N}) &= -9.8 \text{ m/s}^2 \cdot m \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad F_{H1} &= F_g - F_{H2} \\ F_{H1} &= F_{g2} \\ &= (841.7 \text{ kg})(9.8 \text{ m/s}^2) \\ F_{H1} &= 832 \text{ N} \\ &= m \end{aligned}$$

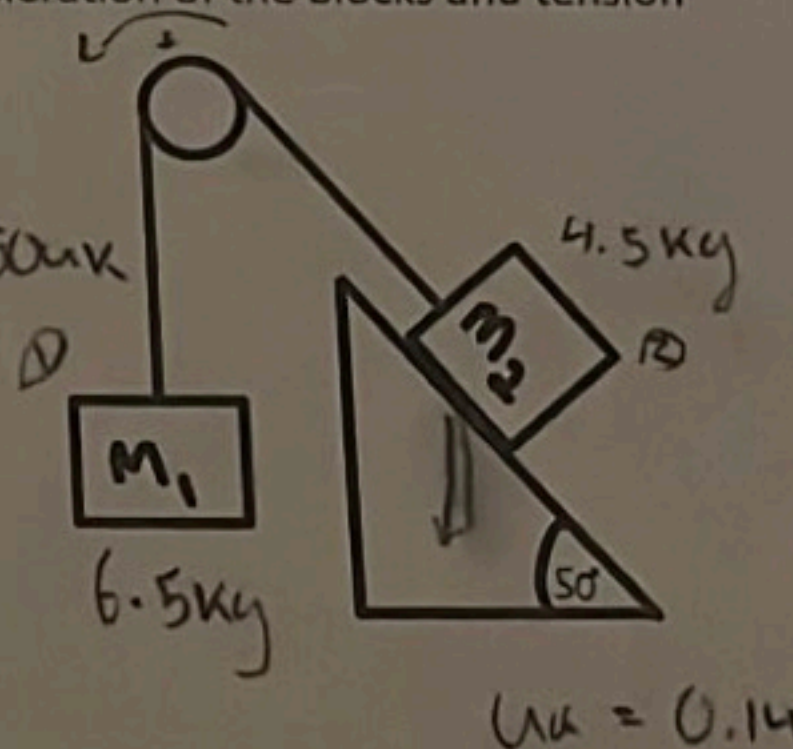
3. Two blocks are connected to each other through a pulley as shown in the diagram below. The block on the ramp has a mass of 4.50 kg and the block on hanging has a mass of 6.5 kg. The coefficient of friction is 0.14. (6 marks)



a) Draw an FBD and FOR for each block and determine the magnitude of acceleration of the blocks and tension of the rope. Show your work. (4 marks)

$$\begin{aligned} F &= m_1 a \\ F &= F_g - F_T \\ &= m_1 g - F_T \\ F &= m_2 a \\ F &= F_T - F_g - F_k \\ &= F_T - m_2 g \sin 50^\circ - m_2 g \cos 50^\circ \mu_k \end{aligned}$$

b) If the string broke, for what minimum value of the coefficient of static friction would block 2 not begin to slide? (2 marks)



None or sep $\textcircled{2}$ start of paper

$$3a) \quad a = 2.36 \text{ m/s}^2 \quad F_T = 480 \text{ N}$$

$$3b) \quad \mu_k = 1.14$$



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Communication (14 mark):

1. Plot a position time graph for the object; make sure your line or curve is clearly shown. (3 marks)

$$A = \frac{1 \cdot 40}{2}$$

$$= 20m$$

$$B = \frac{3 \cdot 60}{2}$$

$$= 90m$$

$$C = 3 \cdot 40$$

$$= 120m$$

$$AB(1-4) : 230m$$

$$D = \frac{5 \cdot 100}{2}$$

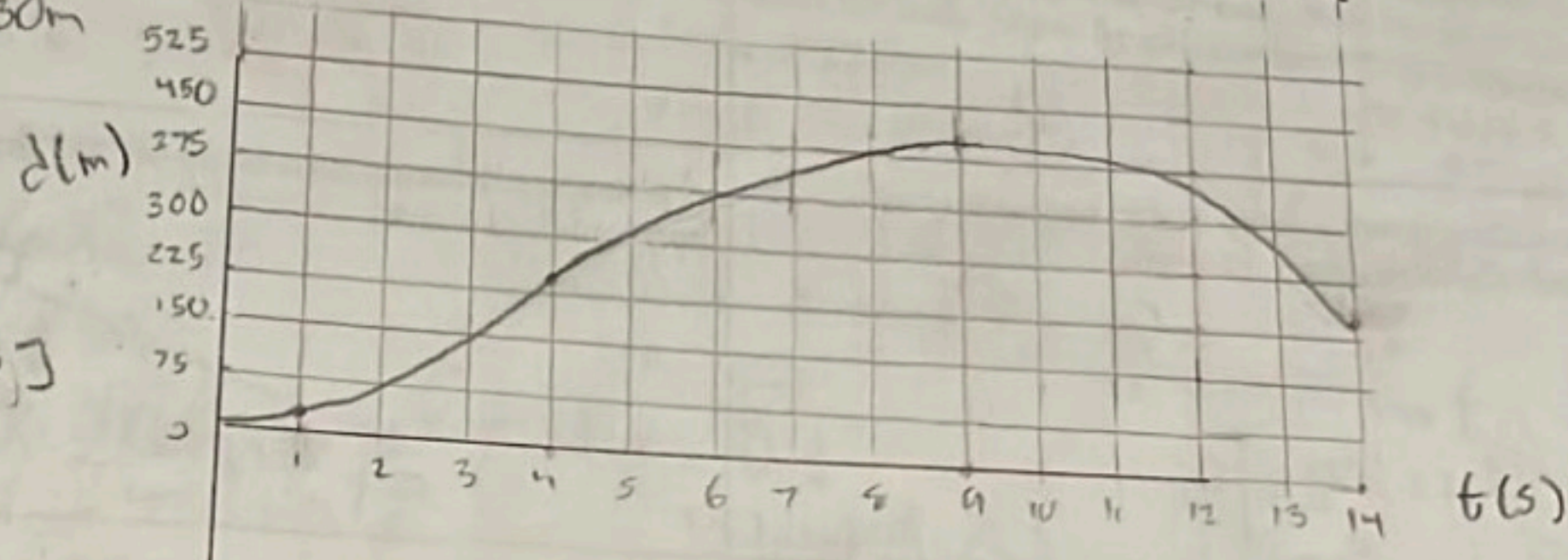
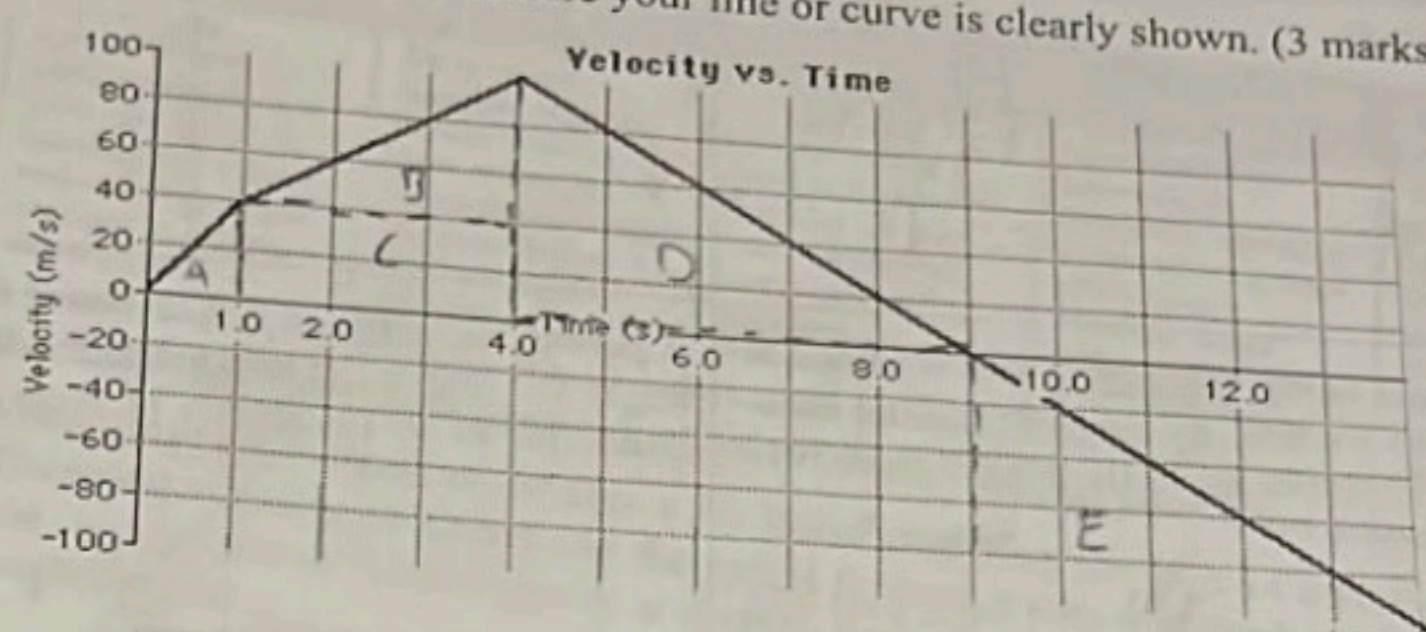
$$= 250$$

$$D = 480m \text{ (up)}$$

$$E = \frac{-100 \cdot 5}{2}$$

$$= 250m \text{ (down)}$$

$$E = 230m \text{ (down)}$$



b) Plot an acceleration time graph for the object. (2 marks)

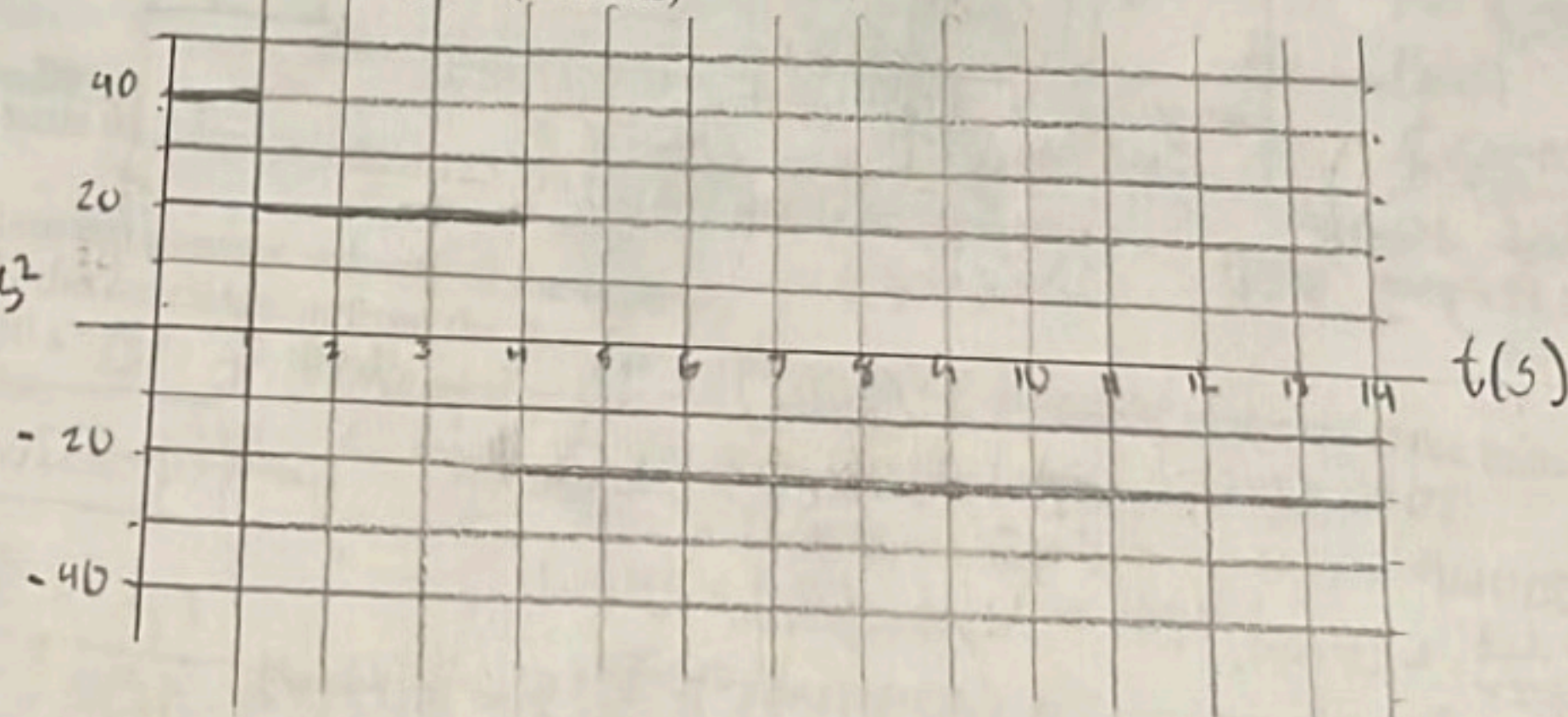
$$A = 40m/s^2$$

$$BC = 20m/s^2$$

$$D = -20m/s^2$$

$$E = -20m/s^2$$

m/s^2



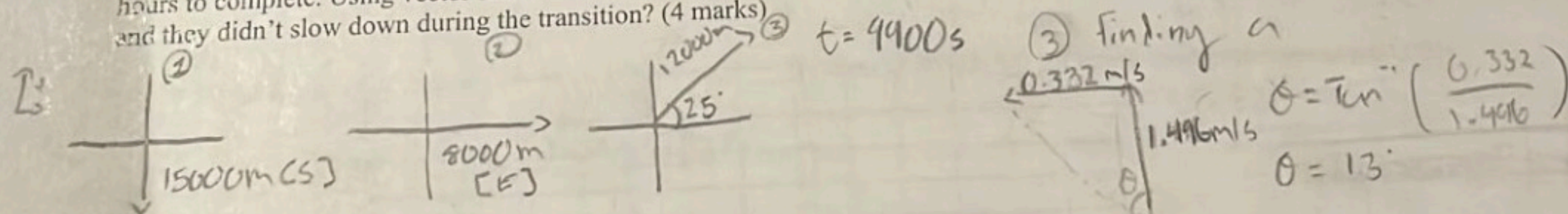
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2. CLEARLY draw an FBD for the following. Be sure to label all appropriate forces. Relative lengths must be obvious! (5 marks)

A person standing in an elevator accelerating downwards (0.5 marks)	A hockey puck moving right on ice with NO friction, after being hit with a hockey stick (0.5 marks)
A box full of textbooks being dragged across the floor at constant velocity, with friction, by a person pulling on a rope at an angle of 23° above the horizontal (1 mark)	A person standing on a skateboard that is flying on an angle 35° above the horizontal through the air, slowing down due to air resistance. (1 mark)
A block sliding down a ramp at 40° below the horizontal. Slowing down. (1 mark)	A plane accelerating as it takes off 55° degrees above the horizontal. (1 mark)

3. A lady rides a camel at a constant speed 15.0 km [S], then 8 km [E] and finally 12.0 km [E 25° N]. The trip takes 2.75 hours to complete. Using vector subtraction, what is the acceleration between legs 2 and 3 of the trip if the transition took 40s and they didn't slow down during the transition? (4 marks)



① Finding avg speed ② comps

$$v = \frac{d}{t}$$

$$= \frac{35000m}{9900s}$$

$$v = 3.54 m/s$$

$$v_y = (3.54 m/s) \sin 25$$

$$= 1.496 m/s [N]$$

$$v_x = v_{3x} - v_{2x}$$

$$= (3.54 m/s) \cos 25 - 3.54 m/s$$

$$= 0.332 m/s [W]$$

$$a = \frac{v}{t}$$

$$= \frac{\sqrt{(0.332 m/s)^2 + (1.496 m/s)^2}}{40s}$$

$$a = 0.038 m/s^2$$

∴ acceleration

between leg 2 & 3 is

$$a = 0.038 m/s^2 [N 13° W]$$



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